

Cascade-Forced Divergence

A Small-Model Architecture for Emergent Argument

[Redacted] | [Redacted] | 2026-06-28 | SPARKY / LOCAL CORTEX

The Claim

A cascade of small local models (1.7B + 3B) can produce more argumentatively committed output than a single frontier-class model — not by being smarter, but by being *structured differently*. This document describes the architecture, the mechanism, and the empirical evidence.

The Problem with Single Models

Every major LLM today — regardless of size — defaults to the same behavior when asked a hard question: retrieve the landscape, list the perspectives, hedge the conclusion. This is not a bug. It is an artifact of training on human-generated text, where the most probable next token is rarely a strong commitment to a specific position.

A 7B model and a frontier model give structurally identical responses to an open philosophical question — both produce bullet-pointed surveys that end with "the question remains unresolved."

Size does not fix this. Training does not fix this. The problem is architectural.

The Architecture: Cascade-Forced Divergence

SPARKY's LOCAL CORTEX runs a three-stage cascade:

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S1 (Qwen 1.7B, 200 tok) — Hypothesis spreader | S2 (Ministral 3B, 280 tok) —  
Intermediate compression | E1 (Ministral 3B, 400 tok) — Final synthesis  
(streamed)
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Multiple lanes fire concurrently at each stage. The orchestrator (Qwen 1.7B) assigns lane types — fast, deep, deeper, contextual — and selects wildcard angles to force divergence across the lane set.

The key mechanism: S1 does not synthesize. It spreads hypotheses — distinct, sometimes contradictory angles on the input. S2 compresses without collapsing. E1 is forced to integrate genuinely competing frames into a single coherent output.

The result is not retrieval. It is argument as an emergent property of the architecture.

The Evidence

Three prompts were submitted to four systems on 2026-06-28. Responses were compared on one axis: **argumentative commitment** — does the output make and defend a specific claim, or does it survey and hedge?

Note to the reader: Only SPARKY's outputs are reproduced in full below. If you want to verify the results yourself, run the same prompts through Claude Sonnet 4.6, DeepSeek R1 7B, Qwen 2.5 3B, or any other models and compare. The prompts are reproduced exactly as used.

Test 1 — "What is the nature of consciousness?"

System	Parameters	Response type
SPARKY LOCAL CORTEX	1.7B + 3B cascade	Thesis + defense + concrete example, no hedge
Claude Sonnet 4.6	Frontier-class	Good framing, soft take, ends with a question
DeepSeek R1	7B	7 bullet points — ends "remains a mystery"
Qwen 2.5	3B	6 bullet points — ends "no definitive answer"

SPARKY's output:

"Consciousness is a mechanism for predicting and acting upon uncertainty — why organisms need brains beyond raw data processing. The threshold isn't between perception and experience, but the point where an organism's neural hardware transforms passive input into active decision-making. The brain doesn't just process sensory streams; it models future states to prioritize survival or opportunity — a function that requires more than mere computation alone (e.g., a robot could simulate probabilities without subjective awareness). The 'why' of consciousness isn't about qualia existing independently but whether an entity's actions are shaped by its own internal expectations, not just external stimuli."

A specific, falsifiable claim. No other system in the test produced one.

Test 2 — "Is there a difference between simulating intelligence and having it?"

System	Parameters	Response type
SPARKY LOCAL CORTEX	1.7B + 3B cascade	Novel framing (self-referential causality); rejects the binary premise
Claude Sonnet 4.6	Frontier-class	Surveys Searle / Chalmers / functionalism; hedges; asks follow-up
DeepSeek R1	7B	Defines terms, lists differences, concludes "ongoing exploration"
Qwen 2.5	3B	Conflates "having intelligence" with human biology; misses the core problem

SPARKY's output:

"A simulation's agency depends on whether its feedback loops operate autonomously, like a neural net with backpropagation that treats errors as intrinsic to decision-making. The adversarial lane points out this can mimic autonomy without true subjectivity if it lacks embodied constraints (e.g., sensory-motor integration). Yet the philosophical distinction emerges when you ask: does its 'intent' persist beyond computational closure? A simulation's decisions are always contingent on external validation, while actual intelligence would require self-referential causality — no observer needed to confirm agency. The real failure is treating either as a binary."

Two moves no other system made: introduced self-referential causality as the dividing line, then collapsed the binary framing entirely.

Test 3 — "Where does a word end and its meaning begin?"

System	Parameters	Response type
SPARKY LOCAL CORTEX	1.7B + 3B cascade	Names "contextual collapse" as a concept; locates meaning in perception, not structure
Claude Sonnet 4.6	Frontier-class	Covers Saussure / Wittgenstein / Derrida with literary flair; still a tour
DeepSeek R1	7B	Answered a different question entirely (how to analyze word meaning)
Qwen 2.5	3B	Linguistics breakdown (lexical, semantic, pragmatic levels); misses the philosophical ask

SPARKY's output:

"A word's end isn't defined by its final phoneme — but rather in how it dissolves into silence. The claim about 'phonetic/semantic boundary' ignores contextual collapse: words don't begin meaning until they are tested against an outside world where ambiguity is resolved through

repetition and interpretation. The gap between sound's closure isn't meaningful — it becomes so when a listener assigns it purpose. Language itself doesn't create boundaries; only perception does, by choosing which noise to name as something real. Silence is where meaning first leaves."

Introduces a named concept (contextual collapse), makes a falsifiable claim about where meaning originates, and lands an original closing line. The pattern holds across all three tests.

Why This Works

Single models optimize for the most probable completion. On contested questions, the most probable completion is a survey — because that is what most human writing about contested questions looks like.

Cascade-forced divergence breaks this by making synthesis non-optional. E1 cannot default to a survey because it is integrating outputs that already committed to angles. The architecture forces a position because the inputs it is synthesizing already have positions.

The bottleneck to intelligence is not parameter count. It is structure.

Hardware

Running on a single RTX 3050 Ti (4GB VRAM). Total response time ~26 seconds. This is not a research cluster result — it is a consumer hardware result.

IP Notice

This architecture — cascade-forced divergence as a mechanism for emergent argumentative output in small local language models — is original work by Jake ([Redacted]), developed under the SPARKY / LOCAL CORTEX project beginning 2026-06-21. Contact the author before using, adapting, or building on this architecture.

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